

Stiven Gjinaj

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Master's graduate in Computer Engineering specializing in AI and Data Analytics. Proven ability to manage and analyze Big Data using Deep Learning models. Experienced as a Freelance Full-stack Developer with a solid foundation in software engineering.

EDUCATION

Politecnico di Torino

Master's Degree in Computer Engineering - AI and Data Analytics

Torino, Italy

March 2024 – July 2026

Alma Mater Studiorum - Università di Bologna

Bachelor's Degree in Computer Engineering and Science

Cesena, Italy

September 2020 – March 2024

EXPERIENCE

Master Thesis Researcher – SentinelPV Project

December 2025 – June 2026

Department of Control and Computer Engineering & Department of Energy

Torino, IT

- Developed Deep Learning models (Transformers and CNNs) for regional energy production reconstruction, analyzing data from sentinel photovoltaic plants.
- Managed and processed a massive 90GB+ dataset provided by Terna (under NDA), integrating grid time-series with meteorological and geographical data.
- Designed a Data Cleaning and Data Integration pipeline to merge heterogeneous sources such as PVGIS and Open-Meteo, ensuring spatial and temporal alignment.
- Optimized large-scale model performance by implementing efficient Python data loading and preprocessing techniques to handle high-dimensional datasets.
- Collaborated with energy sector experts to validate AI model results against real-world dynamics of the Piedmont electricity grid.

Hackathon Participant – TinyHack

October 2025

Organized by FocoosAI, Z-Ant & Arduino

Torino, IT

- Developed an end-to-end Edge AI system for real-time image classification on Arduino Nicla Vision.
- Trained and optimized a computer vision model via the FocoosAI platform, applying quantization techniques to reduce model footprint without compromising accuracy.
- Designed hardware-software integration for on-device image acquisition and inference, communicating results to a cloud infrastructure via REST APIs.
- Built a remote monitoring system by interfacing hardware with a dedicated backend/frontend for real-time classification data visualization.

Freelance Full Stack Developer

February 2025 – September 2025

Fantashow

Remote

- Designed a full-stack platform for online competitions, implementing algorithms for dynamic score calculation and referral systems.
- Built an advanced administration dashboard for secure data management, ensuring system consistency and fault resilience.
- Designed a NoSQL database on Firebase, implementing secure authentication systems and RESTful APIs for data synchronization.
- Integrated the Stripe payment gateway, managing transactional flows via webhooks and implementing recovery procedures for payment failures.

PUBLICATIONS

Sentinel-Conditioned 24-Hour-Ahead Irradiance Forecasting over Distributed Photovoltaic Fleets 2026

- S. Monaco, S. Gjinaj, A. Ciocia, A. C. Marceddu, and E. Arco, "Sentinel-Conditioned 24-Hour-Ahead Irradiance Forecasting over Distributed Photovoltaic Fleets," in *Proceedings of the Workshop on Machine Learning for Energy Management and Sustainable Energy (ML4EMS@ECML PKDD 2026)*, Naples, Italy, 2026.

PROJECTS

- DNA Sequence Prediction via State Space Models | *S5, HPC, BioInformatics* May 2025 – September 2025
- Engineered an S5 State Space Model architecture for genomic sequence prediction from raw Oxford Nanopore Technology signals.
 - Optimized model training on High Performance Computing (HPC) clusters to process large volumes of bioinformatics data efficiently.
 - Achieved 89% accuracy while reducing model size and complexity by 10x compared to the state-of-the-art (91% accuracy).
- CarRental | *React, Spring Boot, PostgreSQL, Kafka, Docker, Prometheus, Grafana* February 2025 – July 2025
- Developed a microservices-based car rental system containerized with Docker, featuring PayPal integration and OpenAPI documentation.
 - Implemented real-time monitoring and observability using Prometheus and Grafana; utilized Apache Kafka for asynchronous inter-service messaging.
- Out of Kilter Algorithm | *Python, Network Optimization* October 2023 – November 2023
- Implemented the primal-dual out-of-kilter algorithm in Python to solve minimum-cost flow problems in transportation networks.
 - Developed dynamic adjustment logic for arcs and node potentials to ensure optimal convergence on complex graphs.
- Streaming Website | *React, Spring Boot, Firebase, Docker, Render* August 2024 – September 2024
- Created a video streaming platform featuring user account management for saving viewing preferences and history.
 - Built a robust Spring Boot backend and deployed the containerized application automatically on Render via Docker.
- Archiumion | *Kotlin, Jetpack Compose, Firebase, Room Database* April 2023 – June 2023
- Developed an Android social network for architects to share and review professional projects, built with Jetpack Compose.
 - Integrated Firebase for authentication and real-time storage, utilizing Room Database for local data persistence.
- Arduino Smart Home | *Arduino, ESP32, Java, Proximity Sensors, Servo Motors* December 2022 – March 2023
- Designed an integrated smart home automation system utilizing Arduino and ESP32 microcontrollers.
 - Developed a Java control dashboard for real-time sensor monitoring and remote device management via Wi-Fi and Bluetooth.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Kotlin, SQL (PostgreSQL), JavaScript, TypeScript, ARM Assembly
AI & Machine Learning: PyTorch, Scikit-learn, Transformers, State Space Models (S5), Computer Vision (OpenCV), Model Quantization, TinyML, Time Series Analysis
Data Engineering & MLOps: Big Data Integration, Data Wrangling (Pandas, NumPy), Wandb, HPC Clusters, Data Cleaning
Cloud & DevOps: Docker, Apache Kafka, Prometheus, Grafana, CI/CD, Microservices, Render
Frameworks & Web: React, Spring Boot, Node.js, Express, Django, Firebase, Supabase, Stripe/PayPal API, OpenAPI/Swagger
Systems & Software: Kernel Development, Multiprogramming, RESTful APIs, Mobile Dev (Jetpack Compose, Kotlin Multiplatform), Agile/Scrum
Soft Skills: Problem Solving (trade-off optimization), Multidisciplinary Teamwork, Working under pressure, Business Requirements Translation